Voice-Based Concept Understanding Analyser

Submitted By

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PROJECT OVERVIEW

Purpose

The Voice-Based Concept Understanding Analyser is an AI-powered educational platform that evaluates students' conceptual understanding through spoken responses. It converts speech into text, analyses semantic similarity, evaluates speech fluency, and provides personalised AI-generated feedback.

Features

- Voice Recording/Upload
- Speech-to-Text Conversion
- Semantic Similarity Analysis
- Speech Fluency Analysis
- AI-Based Personalised Feedback
- PDF Report Generation
- Student Dashboard & History

ARCHITECTURE

Frontend

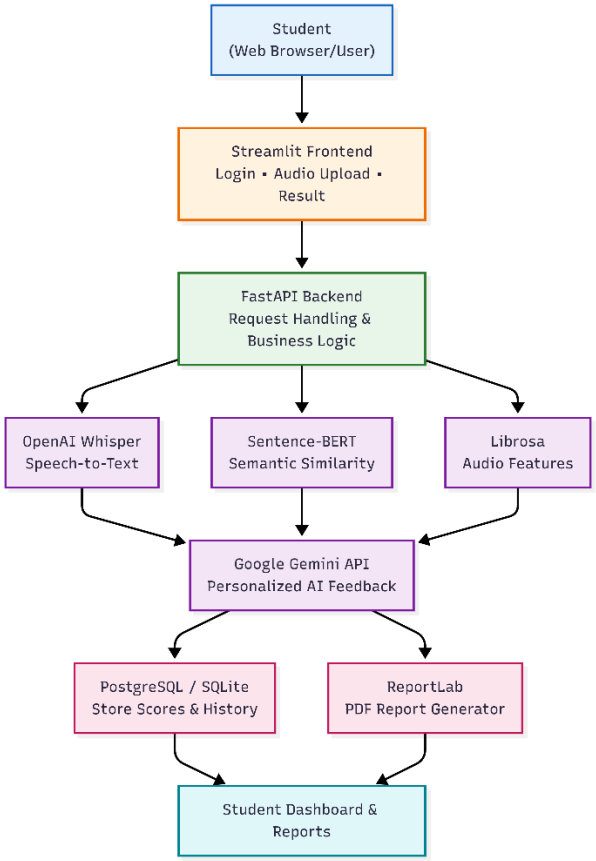
The frontend is developed using **Streamlit**, providing an interactive interface for student login, voice upload, result visualisation, and report download.

Backend

The backend is built using **FastAPI**, which handles API requests, audio processing, AI model integration, score calculation, and report generation.

Database

The system uses **SQLite/PostgreSQL** to store student details, assessment history, scores, transcripts, and generated feedback.



SETUP INSTRUCTIONS

Prerequisites

- Python 3.10+
- pip
- Git
- SQLite/PostgreSQL
- Google Gemini API Key
- OpenAI Whisper
- FFmpeg

Installation

git clone <repository-url>

cd VBCUA

pip install -r requirements.txt

Create a .env file:

GEMINI_API_KEY=your_api_key

DATABASE_URL=your_database_url

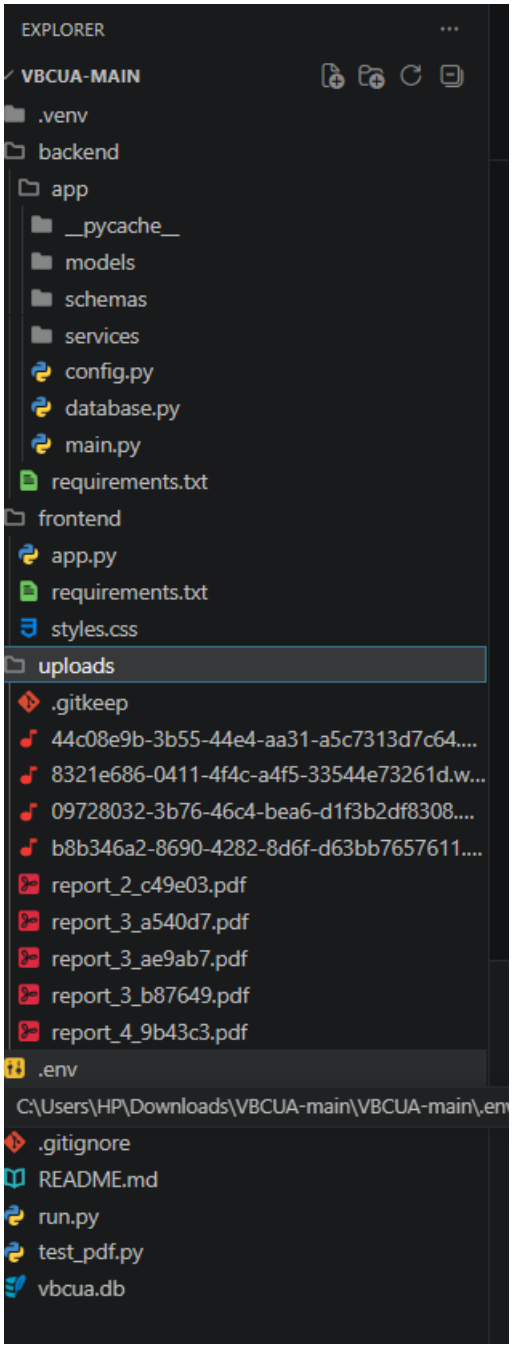
Run the backend:

uvicorn main:app --reload

Run the frontend:

streamlit run app.py

FOLDER STRUCTURE



- **backend/** – Contains the FastAPI backend, business logic, API endpoints, database configuration, and AI processing modules.
- **app/models/** – Defines database models used by the application.
- **app/schemas/** – Contains request and response validation schemas.
- **app/services/** – Implements speech processing, semantic analysis, AI feedback, and report generation services.
- **config.py** – Stores application configuration settings.
- **database.py** – Handles database connection and initialization.
- **main.py** – Main entry point for the FastAPI server.
- **frontend/** – Contains the Streamlit user interface.
- **app.py** – Main Streamlit application.
- **styles.css** – Custom styles for the frontend.
- **uploads/** – Stores uploaded audio recordings and generated PDF reports.
- **.env** – Stores environment variables such as API keys.
- **README.md** – Project documentation and setup instructions.
- **run.py** – Script to launch the application.
- **test_pdf.py** – Used to test PDF report generation.
- **Vbcua.db** – SQLite database storing student information, assessment results, and history.

RUNNING THE APPLICATION

Backend

uvicorn main:app --reload

Frontend

streamlit run app.py

API DOCUMENTATION

Endpoint	Method	Description
/register	POST	Register a new student
/login	POST	Student login
/upload	POST	Upload voice recording
/transcribe	POST	Convert speech to text
/evaluate	POST	Evaluate conceptual understanding
/feedback	GET	Generate AI feedback
/report	GET	Download PDF report

USER INTERFACE

The application includes the following screens:

- Dashboard
- Voice Upload Page
- Evaluation Results
- AI Feedback Screen
- PDF Report Download

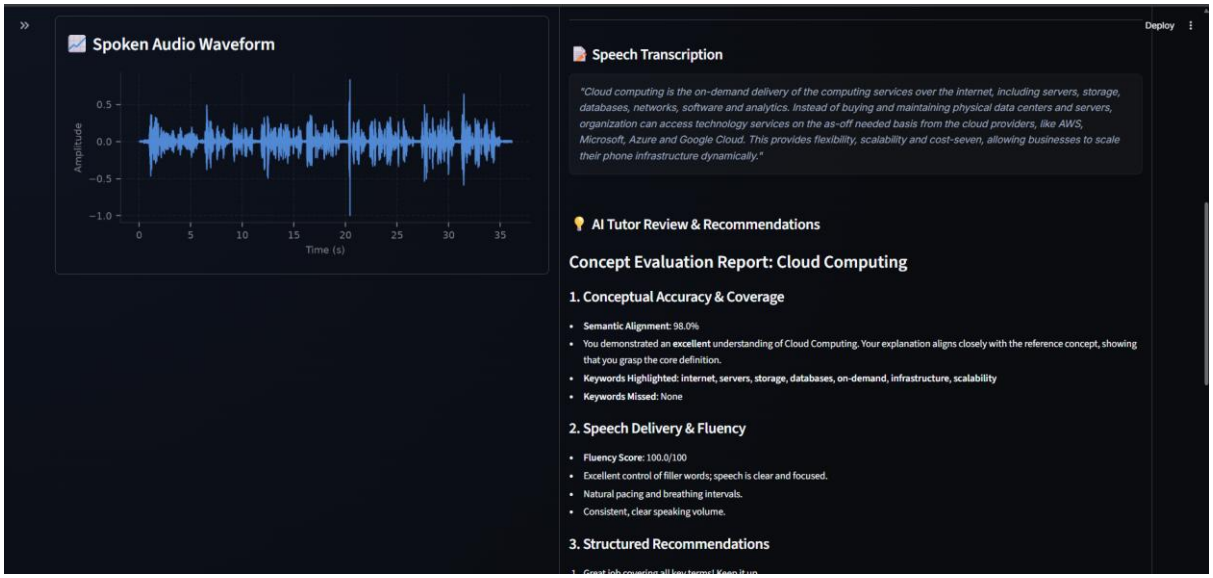


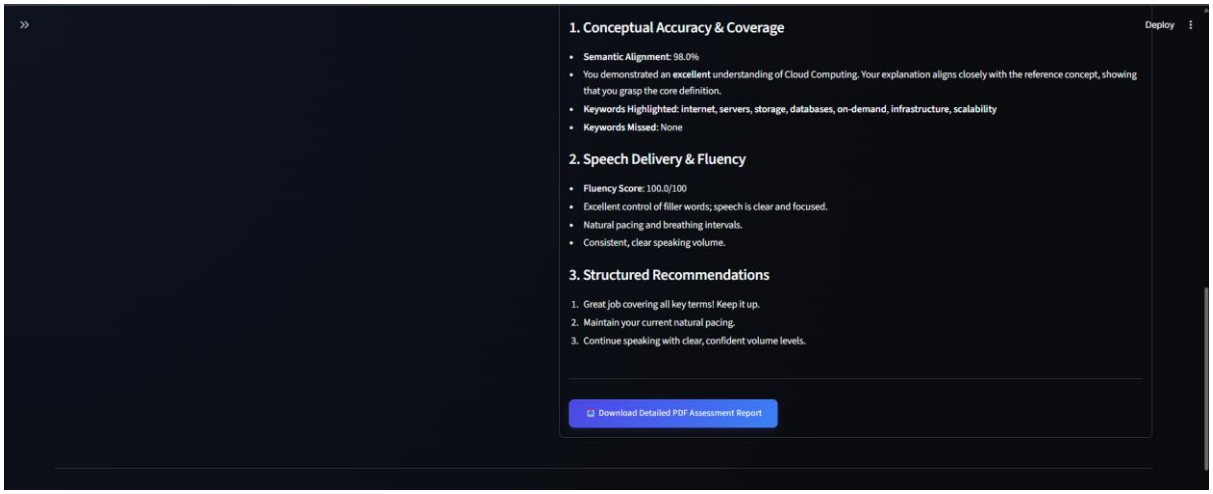
TESTING

The project was tested using Functional Testing, Performance Testing, and User Acceptance Testing (UAT). All core functionalities including login, voice upload, speech recognition, semantic analysis, AI feedback, and PDF generation were verified successfully.

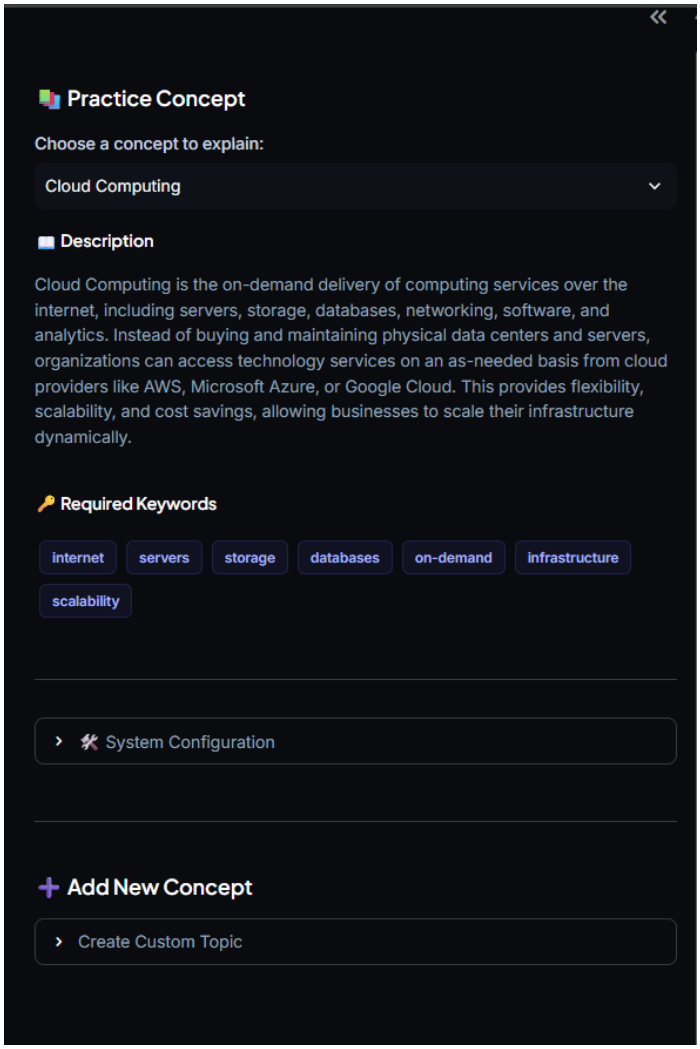
RESULTS


Deploy
VBCUA Assessment Dashboard
AI-Powered Semantic & Speech Fluency Concept Evaluation
Spoken Explanation Input
Explain the selected concept out loud. You can record directly using your microphone or upload an audio file.
Option 1: Record live speech
Click to start recording your voice
00:36
0:00 / 0:36
Option 2: Upload audio file (.wav, .mp3, .m4a)
Upload audio explanation
Upload
200MB per file • WAV, MP3, M4A
Analyze Explanation
Analysis Results: Cloud Computing
OVERALL SCORE
99.0%
Strong Understanding
SEMANTIC MATCH
98.0%
Sentence-BERT Match
SPEECH FLUENCY
94%
P: 8% | F: 1
Comprehension Accuracy & Speech Mastery
0
20
40
60
80
100
99
Core Keyword Coverage Check
✓ internet
✓ servers
✓ storage
✓ databases
✓ on-demand
✓ infrastructure
✓ scalability


Deploy
Spoken Audio Waveform
Amplitude
0.5
0.0
−0.5
−1.0
0
5
10
15
20
25
30
35
Time (s)
Speech Transcription
"Cloud computing is the on-demand delivery of the computing services over the internet, including servers, storage, databases, networks, software and analytics. Instead of buying and maintaining physical data centers and servers, organization can access technology services on the as-off needed basis from the cloud providers, like AWS, Microsoft, Azure and Google Cloud. This provides flexibility, scalability and cost-seven, allowing businesses to scale their phone infrastructure dynamically."
AI Tutor Review & Recommendations
Concept Evaluation Report: Cloud Computing
1. Conceptual Accuracy & Coverage
Semantic Alignment: 98.0%
You demonstrated an excellent understanding of Cloud Computing. Your explanation aligns closely with the reference concept, showing that you grasp the core definition.
Keywords Highlighted: internet, servers, storage, databases, on-demand, infrastructure, scalability
Keywords Missed: None
2. Speech Delivery & Fluency
Fluency Score: 100.0/100
Excellent control of filler words; speech is clear and focused.
Natural pacing and breathing intervals.
Consistent, clear speaking volume.
3. Structured Recommendations


Deploy
1. Conceptual Accuracy & Coverage
Semantic Alignment: 98.0%
You demonstrated an excellent understanding of Cloud Computing. Your explanation aligns closely with the reference concept, showing that you grasp the core definition.
Keywords Highlighted: internet, servers, storage, databases, on-demand, infrastructure, scalability
Keywords Missed: None
2. Speech Delivery & Fluency
Fluency Score: 100.0/100
Excellent control of filler words; speech is clear and focused.
Natural pacing and breathing intervals.
Consistent, clear speaking volume.
3. Structured Recommendations
1. Great job covering all key terms! Keep it up.
2. Maintain your current natural pacing.
3. Continue speaking with clear, confident volume levels.
Download Detailed PDF Assessment Report


Evaluation Performance History
DATE
CONCEPT
OVERALL SCORE
GRADE
DURATION
FILLER WORDS
INSPECT
Jul 19, 2026 20:30
Cloud Computing
99.0%
Strong Understanding
36.1s
1
View
Jul 19, 2026 19:48
Cloud Computing
99.0%
Strong Understanding
36.1s
1
View
Jul 19, 2026 13:42
Database Management System (DBMS)
82.0%
Strong Understanding
34.9s
1
View
Jul 19, 2026 13:22
college
64.0%
Moderate Understanding
21.9s
1
View
Jul 19, 2026 13:22
Cloud Computing
49.4%
Poor Understanding
5.8s
0
View
Select a past attempt to inspect:
Attempt #5: Cloud Computing - Jul 19, 2026 20:30 (Score: 99.0%)
Load Attempt


Practice Concept
Choose a concept to explain:
Cloud Computing
Description
Cloud Computing is the on-demand delivery of computing services over the internet, including servers, storage, databases, networking, software, and analytics. Instead of buying and maintaining physical data centers and servers, organizations can access technology services on an as-needed basis from cloud providers like AWS, Microsoft Azure, or Google Cloud. This provides flexibility, scalability, and cost savings, allowing businesses to scale their infrastructure dynamically.
Required Keywords
internet
servers
storage
databases
on-demand
infrastructure
scalability
System Configuration
Add New Concept
Create Custom Topic